AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T316
Advanced Networking

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 3/Dec. /2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: True/False Questions (10 marks)

Question 1: Put True/False for each of the following definitions (1 mark/Q)

1. IntServ (Integrated Services) QoS model requires per-flow state information to be maintained at each router along the path. (.....)
2. In MPLS networks, the Time-to-Live (TTL) field in the label header is used for loop detection and prevention. (.....)
3. BGP route reflectors eliminate the need for full mesh connectivity between all IBGP routers within an autonomous system. (.....)
4. SONET/SDH uses synchronous multiplexing, where all tributary signals are synchronized to a common clock reference. (.....)
5. Elastic traffic refers to applications that can adapt to available bandwidth and tolerate variations in delay and throughput. (.....)
6. Graded-index multimode fibers have a uniform refractive index throughout the core, like step-index fibers. (.....)
7. BGP path vector protocol prevents routing loops by including the complete AS path in route advertisements. (.....)
8. The pointer mechanism in SONET/SDH allows for flexible payload positioning within the frame structure to accommodate timing variations. (.....)
9. Chromatic dispersion affects only multimode fibers and does not impact single-mode fiber transmission. (.....)
10. Jitter refers to the fixed time delay experienced by all packets in a single data flow. (.....)

PART 2: Short Answer Questions (20 Marks)

Question 1: (10 marks)

A) Explain the key differences between IntServ and DiffServ QoS architectures, highlighting three main distinctions. (6 Marks)

B) A DS3 system multiplexes 672 voice channels, each digitized at 8 bits per sample every 125 μs. Calculate the total data rate of the DS3 signal. (4 Marks)

Question 2: (10 marks)

A) In MPLS networks, explain the purpose and function of the Label Distribution Protocol (LDP). How does it establish label-switched paths? (4 marks)

B) Describe the three main fields in an MPLS label header and explain the significance of each field. (6 marks)

PART 3: Long Answer Questions (30 Marks)

Question 1: (15 marks)

A) In MPLS networks, explain the difference between label stacking and label swapping. Under what circumstances would an LSR perform each operation? (7 marks)

B) What is the difference between Link State routing protocols and Distance Vector routing protocols? Define each one and give examples (8 marks)

Question 2: (15 marks)

A) Define Attenuation in optical fibers. Name and briefly explain two main causes of attenuation. (7 marks)

B) A fiber of 5 km in length has Pin = 250 μW and Pout = 200 μW. Find the loss in dB/km. (8 marks)

End of questions

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